جامعة خليفة
Khalifa University

Job Description

1. General Details

Job Code	
Job Title	Scientific Computing Support , Senior Specialist
Reporting Line	Manager, Scientiific Computing Support
Department	Research Computing

2. Job Purpose

Scientific Computing Support, Senior Specialist is responsible for assisting researchers with running their codes and applications on the research computing facilities, training them on the correct usage of those resources, assisting with analysis of performance, code optimization, and parallel scaling of scientific applications, as well as the implementation of parallel algorithms based on C/C++, Fortran, Matlab, Python, MPI, OpenMP, CUDA.

3. Key Roles & Responsibilities

Strategic Responsibilities

N/A

Operational Responsibilities

- Support researchers by understanding their research computational needs and recommend suitable applications, such as Ansys Fluent, Lumerical, Abaqus, Comsol, OpenFoam…, and other research computing resources available at the University.
- Ensure the resolution of faculty and student computing resource issues and provide programming consultancy, code optimization, and resolve users' programming issues (based on shell scripting, C/C++, Matlab, Python, MPI, OpenMP, and CUDA).
- Assist researchers with grant proposals that have a strong High-Performance Computing (HPC) component to describe the utilization of HPC resources in their research.
- Work with the HPC sysadmins to build, install, maintain, and test scientific applications on the HPC clusters and VMs.
- Develop awareness and training programs for students, faculty, and staff concerning research computing techniques, services, and facilities, including HPC application training to help users parallelize, debug, and optimize their applications.
- Conduct regular training, workshops, and tutorials, as well as one-on-one training for new and experienced users.
- Liaise with research teams to continuously discuss emerging issues and implement service enhancements.
- Assist sysadmins with testing upgrades to the systems and new applications.
- Assist with the development of related outreach programs and projects at the University.
- Maintain and update the Research Computing knowledge base.

This document is the Property of Khalifa University of Science and Technology, and should not be distributed, used and given to any outside party without prior authorizations

- Improve existing tools and develop new monitoring and reporting tools used to generate statistical reports of the utilization of the computational resources.
- Suggest and implement improvements to policies, procedures, and configuration of research computing resources.
- Assist in monitoring the impact of the research computing facilities and services on the university's scientific output
- Adhere to the University's information security and confidentiality policies and procedures, and report breaches or other security risks accordingly
- Coordinate with other departments to facilitate the accomplishment of tasks and responsibilities, as and when needed
- Perform any other tasks assigned by the Line Manager

Supervisory Responsibilities

N/A

4. Qualifications & Experience

Required Qualifications

- Bachelor's Degree or equivalent in a related scientific field (astrophysics, life sciences, computational chemistry, as well as computational sciences, mathematics, physics etc.).
- Good programming skills in at least one high-level programming language (C, C++, Fortran) and at least one scripting language (bash, Python, Matlab, etc.).

Other Requirements

- Experience in Parallel Programming (e.g. MPI, OpenMP, CUDA) and performance analysis, libraries and tools.
- Knowledge of HPC environments and schedulers (LSF, PBS, Slurm, modules, lmod…).
- Experience with compilation techniques and usage methodologies of a range of scientific, technical, and research focused applications.
- Experience with scientific applications, e.g. Anys Fluent, Comsol, Abaqus, Lumerical…, is desirable.
- Proven experience providing technical support to a user base with a wide range of experience and skill levels.
- Experience working in a higher education environment would be desirable.
- Good presentation skills.
- Good communication and interpersonal skills.
- Ability to work well both individually and as part of a team.

This document is the Property of Khalifa University of Science and Technology, and should not be distributed, used and given to any outside party without prior authorizations

Job Description

Required Experience	
Expats: A minimum 7-9 years of relevant experience.	UAE Nationals: A minimum of 3-5 years of relevant experience

This document is the Property of Khalifa University of Science and Technology, and should not be distributed, used and given to any outside party without prior authorizations